

Ch4_6

$$\textcircled{a} \Pr(Y=1 | x_1=40, x_2=3.5) = \frac{\beta_0 + \beta_1 x_1 + \beta_2 x_2}{1 + \exp(-\beta_0 + 0.05x_1 + 1x_2)} = \frac{1}{1 + \exp(0.5)} = 0.3775 = \underline{37.8\%}$$

$$\textcircled{b} \log\left(\frac{0.5}{1-0.5}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 = -6 + 0.05x + 3.5$$

$$\log 1 = -2.5 + 0.05x$$

$$2.5 = 0.05x \quad \underline{x=50}$$

Ch4_9

$$\textcircled{a} \text{odds} = \frac{P}{1-P} = 0.37$$

$$\Rightarrow 1.37 P = 0.37$$

$$\Rightarrow P = \frac{0.37}{1.37} = 0.27 \Rightarrow \underline{27\%}$$

$$\textcircled{b} \text{odds} = \frac{0.16}{1-0.16} = \frac{0.16}{0.84} = \underline{0.19}$$

Ch4_12

$$\textcircled{a} \text{log-odds} = \log\left(\frac{\hat{P}_r(Y=\text{orange} | x=x)}{\hat{P}_r(Y=\text{apple} | x=x)}\right) \stackrel{\text{Me}}{=} \log\left(\frac{\frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 x)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 x)}}{\frac{1}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 x)}}\right) = \log(\exp(\hat{\beta}_0 + \hat{\beta}_1 x))$$

$$= \underline{\hat{\beta}_0 + \hat{\beta}_1 x}$$

$$\textcircled{b} \text{log-odds} = \log\left(\frac{\hat{P}_r(Y=\text{orange} | x=x)}{\hat{P}_r(Y=\text{apple} | x=x)}\right) \stackrel{\text{friend}}{=} \log\left(\frac{\frac{\exp(\hat{\alpha}_{\text{orange}0} + \hat{\alpha}_{\text{orange}1} x)}{\exp(\hat{\alpha}_{\text{orange}0} + \hat{\alpha}_{\text{orange}1} x) + \exp(\hat{\alpha}_{\text{apple}0} + \hat{\alpha}_{\text{apple}1} x)}}{\frac{\exp(\hat{\alpha}_{\text{apple}0} + \hat{\alpha}_{\text{apple}1} x)}{\exp(\hat{\alpha}_{\text{orange}0} + \hat{\alpha}_{\text{orange}1} x) + \exp(\hat{\alpha}_{\text{apple}0} + \hat{\alpha}_{\text{apple}1} x)}}\right)$$

$$= \log\left(\frac{\exp(\hat{\alpha}_{\text{orange}0} + \hat{\alpha}_{\text{orange}1} x)}{\exp(\hat{\alpha}_{\text{apple}0} + \hat{\alpha}_{\text{apple}1} x)}\right) = \hat{\alpha}_{\text{orange}0} + \hat{\alpha}_{\text{orange}1} x - (\hat{\alpha}_{\text{apple}0} + \hat{\alpha}_{\text{apple}1} x)$$

$$= (\hat{\alpha}_{\text{orange}0} - \hat{\alpha}_{\text{apple}0}) + (\hat{\alpha}_{\text{orange}1} - \hat{\alpha}_{\text{apple}1}) x$$

$$\textcircled{c} \hat{\beta}_0 + \hat{\beta}_1 x = (\hat{\alpha}_{\text{orange}0} - \hat{\alpha}_{\text{apple}0}) + (\hat{\alpha}_{\text{orange}1} - \hat{\alpha}_{\text{apple}1}) x = 2 + (-1)x$$

$$\Rightarrow (\hat{\alpha}_{\text{orange}0} - \hat{\alpha}_{\text{apple}0}) = 2$$

$$(\hat{\alpha}_{\text{orange}1} - \hat{\alpha}_{\text{apple}1}) = -1$$

$$\textcircled{d} \therefore \hat{\beta}_0 + \hat{\beta}_1 x = (\hat{\alpha}_{\text{orange}0} - \hat{\alpha}_{\text{apple}0}) + (\hat{\alpha}_{\text{orange}1} - \hat{\alpha}_{\text{apple}1}) x = (1.2 - 3) + (-2 - 0.6)x$$

$$\Rightarrow \hat{\beta}_0 = -1.8$$

$$\underline{\hat{\beta}_1 = -2.6}$$

\textcircled{e} For Binary Classification, this two model are equivalent, they predicted $\Pr(Y=\text{orange} | x) > 0.5$ or not, so two models agree 100%